Conduction vs Insulation by RealQM

and the Missing Stiffness

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Abstract

Can RealQM, as a new alternative to textbook quantum mechanics, explain the difference between a conductor and an insulator? RealQM represents an N-electron system by N non-negative unit charge densities in ordinary three-dimensional space, one per electron, meeting at free boundaries and minimising the total energy — Coulomb interaction plus kinetic energy, with no other force and nothing fitted. It is a computable, parameter-free model at polynomial cost.

The difficulty is one of scale. Atomic energies are tens of electron-volts; semiconductor activation is 50–500 meV; metallic conduction sets in near 20 meV. Against the total energy of the system computed here those are 2×10^{-3}, 5×10^{-5} and 5×10^{-6} — so whether a material conducts is decided by a difference five or six orders of magnitude below the number a calculation returns, and must survive being extracted from it. Insulation is then not a separate phenomenon but the absence of that difference.

Two models, two verdicts. An **atomic chain** — the valence electron occupying its own kernel's region — gives an extensive polarisability, $\alpha = 27.4N$ flat to 1% per atom, and a corrugation of order an electron-volt for the partition to slide through: the atomic scale, and an insulator. An **ion–electron cable** — ion cores along an axis with the carriers standing off them, which is how conducting matter is built — gives ≈240 meV, the semiconductor range, where transport is thermally activated hopping; and with the carrier stiffness raised above the value atomic standardisation sets, ≈24 meV, the metallic onset, where the pinning no longer holds and charge is free to move. Not as a metal, since non-overlap leaves no extended states to fill into bands, but as a charge lattice sliding collectively over the ions.

Conduction at 20 meV is reached by raising the coefficient of the kinetic energy from the $\frac{1}{2}$ of atomic standardisation to 1, which compensates for a free interface that resists neither compression nor modulation. The corrugation falls elevenfold. Why an extended object is entitled to a coefficient an atom's spectrum does not settle is argued in the text.

The corrugation is the activation energy E_a. It fixes the exponential in the hopping conductivity, $\sigma \propto \nu \exp(-E_a/kT)$ with kT the thermal energy, so 240 against 24 meV is four orders of magnitude in σ at room temperature. The prefactor ν is an attempt frequency, a lattice quantity this framework can also supply since it moves nuclei under classical equations of motion — though it is not computed here. Metallic conduction is different: its rate needs a relaxation time from electron–phonon scattering, and that stays outside.

1 The question, and the framework it is asked of

This article shows what RealQM, as a new alternative to textbook quantum mechanics, offers as explanation of the difference between conductors and insulators — given only the charges present and their Coulomb interaction, with nothing further supplied. The answer turns out to depend on what the carrier is asked to be, and it is reached by measuring two models.

RealQM [1, 2] represents an N-electron system by N non-negative densities ψ_i, each carrying unit charge on its own domain Ω_i, the domains non-overlapping and their interfaces free to move to where the energy is least. The energy minimised is Coulomb's:

$$E[\{\psi_i\},\{\Omega_i\}] = \sum_i \int_{\Omega_i} \Big(\tfrac{1}{2}|\nabla\psi_i|^2 - \sum_a \frac{Z_a}{|x-x_a|}\psi_i^2\Big)dx + \tfrac{1}{2}\sum_{i\neq j}\iint \frac{\psi_i^2(x)\psi_j^2(y)}{|x-y|}\,dx\,dy, \qquad (1)$$

subject to $\int_{\Omega_i}\psi_i^2 = 1$. There is no wavefunction in $3N$ dimensions, no antisymmetry and no self-interaction; the role played by the exclusion principle is played by the requirement that domains do not overlap. And — of particular relevance here — there is no smeared continuum anywhere in the construction: everything in it is a discrete unit charge occupying a region of ordinary space.

Minimising (1) gives stationarity conditions that take the form of Schrödinger's equation — N of them in three dimensions, coupled through the Coulomb potential and through the free boundaries — together with a condition fixing where each domain ends. The minimisation runs over the partition $\{\Omega_i\}$ as well as over the densities, so the domain edges are unknowns of the problem and not a discretisation of it. The physical content is kinetic energy and the Coulomb interaction, with no other force and nothing fitted.

A two-component system of ions and electrons is therefore not an idealisation the theory must be adapted to describe. It is what the theory is already made of, in a new arrangement.

Every previous test of this construction — ionisation energies across the periodic table, molecular binding, nuclear binding by dual Coulomb confinement [3] — has been on *bound* matter, with an attractor always present. A plasma is the first case in which large numbers of charges of both signs are free to arrange themselves, and it is a fair question what the framework then gives.

The difficulty is one of scale, and it falls on the model as much as on the arithmetic. Atomic energies are tens of electron-volts. Activation energies in the semiconductors and polaronic oxides this framework can hope to describe are 50–500 meV. Metallic conduction sets in near 20 meV, which is thermal energy at room temperature. Against the total energy of a system large enough for the question to be posed — some 4000 eV for the sixteen-site wire computed in §9 — those are 2×10^{-3}, 5×10^{-5} and 5×10^{-6} respectively. So whether a material conducts is settled by a difference five or six orders of magnitude below the number the calculation returns, and the whole difficulty is extracting it without the error swamping it. Two things follow, and both shape the article. The *computation* must produce a difference of that size out of totals five or six orders larger, which is why the controls here occupy as much space as the results and why more than one measurement has been withdrawn. And the *model* must be right at that resolution: a description adequate for binding energies, where an error of a per cent is invisible, can be hopeless for conduction, where a per cent of the total is a hundred times the quantity in question. Insulation, on this view, is not a separate phenomenon requiring its own account: it is the absence of that difference.

The question is not posed here as a contest. Band theory answers it already, at the size of a crystal and correctly, by filling the bands of a periodic potential and counting — supplying that potential rather than deriving it, as density functional theory supplies a functional. What is asked here is different and narrower: whether *this* construction, given the charges and their Coulomb interaction and nothing else, arrives at the distinction by its own route. One framework, one question, and an answer that stands or falls on its own terms.

The whole of the experiment, said in advance. It is two measurements on one chain of seven model atoms carrying one valence electron each, and the reader is owed that before the argument that surrounds them. *First:* put the chain in a uniform field, hold the partition fixed, and read the induced dipole — repeated at three chain lengths, this gives α and how it scales. *Second:* apply no field at all; displace the whole partition rigidly by u and record the energy — this gives the periodic curve $E(u)$, whose amplitude is the corrugation and whose steepest slope is the field at which the pattern would begin to run. Everything else is control: the same converged states reassembled under three boundary conditions, a range of spacings and fillings, a mesh ladder, and the requirement that a shift by one full spacing return the same energy. The article is long because the arguments are, not because the apparatus is.

2 The extension, and the object it is made for

In this article we extend RealQM, developed for atoms and molecules, to **two-component systems**: ions and electrons, both charged, both free to move, interacting by Coulomb forces and nothing else. A metal is such a system, and the question pursued here is the elementary one such a system poses — whether it *conducts*, and if so what carries the charge. The same construction covers a stellar interior and the recombination era of a Coulomb cosmology, treated as a classical plasma.

There the one exact result is the **plasma frequency**, and it is worth saying what it is. Displace the whole electron population rigidly by u against its neutralising ion background: charge neu per unit area is exposed at the two faces, the field between them is $4\pi neu$, and the restoring force per electron is $-4\pi ne^2u$. The population therefore oscillates at $\omega_p^2 = 4\pi ne^2/m$, or $4\pi n$ in atomic units, and the construction returns it exactly.

That is a check on the Coulomb bookkeeping of a two-component system, not evidence for the quantum structure: the derivation is electrostatics and Newton's law, and any theory with the same electrostatics returns the same number. It is quoted here to establish that the extension to ions and electrons is set up correctly before anything harder is asked of it. The quantities that *would* test the quantum structure — the screening length and the dispersion of that oscillation — are where the construction is found wanting, in §5.

3 What conduction asks of a theory

Before asking whether this construction conducts it is worth being exact about what the question demands, because the obvious standard is one that no framework meets.

Conduction is not derived from Schrödinger's equation. The equation is unitary and reversible while resistance is dissipative, and a closed quantum system in a perfect periodic potential, placed in a constant field, reaches no steady current at all: k runs to the zone boundary, Bragg-reflects and returns, so the system rings — *Bloch oscillations* — rather than conducts. A steady current requires irreversibility, and every treatment imports it rather than deriving it: a collision term in Boltzmann transport, a partial trace in an open-system treatment, thermalising reservoirs in Landauer's, τ as an input in $\sigma = ne^2\tau/m$.

What it does supply is the states. Bloch's theorem gives extended solutions in a periodic potential, and Pauli filling then gives the metal–insulator *classification* directly: a partly filled band conducts, a filled band with a gap does not. That is the standing of the question below — whether this framework possesses states of the kind that fill.

So the honest question comes in two tiers, and only the first is a fair test of a structural theory:

tier	status
the classification: which materials conduct	from bands and filling; *this is the fair bar*
the value of σ	needs added statistical machinery in every framework

This article asks the first question of RealQM. It does not ask the second, and no comparison here should be read as faulting the construction for failing to supply a conductivity, since nothing supplies one unaided. What it does examine is whether the framework can reach the classification at all — whether it possesses states of the kind that fill into bands — and the answer, developed below, is that it does not, for reasons that are structural and that the framework's own postulate makes explicit.

Which clock is missing. The first tier needs no time — that is Kohn's point, and it is what lets a static calculation classify. The second does, and the objection to be met is that a minimisation could hardly report a current whatever it was applied to, copper included. So it is worth being exact about which time is absent rather than calling the framework static. A wire contains three time scales, twenty orders of magnitude apart, and the construction lacks them for three different reasons:

what moves	its scale	why this framework does not have it
the signal, hence the energy	$\sim c$	(1) is instantaneous: no retardation, no Maxwell
the carriers	mm/s drift	this *is* σ: inserting it assumes the answer
the electron state	$\hbar/E_h \approx 24$ as	a real ψ has no phase to evolve

Only the last is intrinsic — c arrives by coupling to Maxwell and drift is a transport coefficient — but restoring it does not by itself produce a conductivity. Attosecond variation is far below the resolution of any conduction measurement: scattering times are femtoseconds to picoseconds and a conductivity is an average over many of them, so the phase evolution is integrated out before an ammeter sees anything. What it supplies is not the rate but the *object* — a current, $J \propto \Im(\psi^* \nabla \psi)$, which is identically zero for a real density and cannot be made nonzero by any refinement of a minimisation. What is absent is a *velocity* variable, and that is a weaker requirement than a complex amplitude: in Madelung's hydrodynamic form [4] the same content is carried by a real density together with a real velocity field, with $J = \rho v$ and a continuity equation in place of the phase. The vanishing of $\Im(\psi^* \nabla \psi)$ is therefore a statement about a minimisation, which has no velocity in it at all, and not about real densities, which can perfectly well flow. The rate still comes from the dissipation of the table above.

These two absences are one. A real non-negative density has no phase, and the same lack makes the theory a minimisation rather than an evolution *and* makes the current within a domain vanish identically. That has a positive consequence, and it is the one this article follows: if charge cannot flow inside a domain, the only object left that can carry it is the boundary between domains. It also fixes how the question must be put. A framework with a clock answers "does it conduct" by driving the system and reading a current; this one cannot, so the question is recast as a static one — the maximum gradient of $E(u)$, and the scaling of α with N — which is what §7 and §8 do. The missing clock does not supply the verdict; it determines the form in which a verdict is available at all.

And a pseudo-time is not a clock. There is a time parameter in the solver, and it is worth saying why it is not this one. The fields relax in *imaginary* time, $\tau = it$, under which $e^{-H\tau}$ damps every excited component as $e^{-(E_n - E_0)\tau}$: a filter that finds the ground state, not a dynamics that evolves a state, and the oscillation is not lost by accident but removed on purpose. Charge conservation goes with it. In real time $\partial_t \rho + \nabla \cdot J = 0$ holds exactly; continued to imaginary time the same equation becomes a diffusion with a source, with no flux to conserve and the norm restored by a renormalisation that is global. The relaxation is therefore not an underpowered dynamics that a smaller step would repair — it is the branch of the equation from which the current has been removed.

What does conserve charge locally, in the scheme as it stands, is the interface rule: cells change ownership between neighbouring domains by the local density balance $(\rho_i - \rho_j)/(\rho_i + \rho_j)$, so what leaves one domain arrives in the next. That is a genuine local flux, and it is the reason the boundary and not the electron is the carrier here. It is also where a time-dependent formulation should begin — a continuity law with the boundary velocity as the transport variable — rather than in promoting a descent that was never a conservation law. The descent itself can be promoted in exactly two ways, neither of which answers the question. Overdamped, $\partial_t \Omega = -M\, \delta E/\delta \Omega$, gives a boundary velocity proportional to the driving force — Ohm's law with the correct structure, except that the mobility M is put in by hand and is the conductivity up to geometry. Inertial, giving the moving interface a mass, gives oscillation: the chain rings, as any closed system does, and carries no steady current. The descent rate is a numerical parameter with no calibration to anything physical, and each way of calibrating it either inserts the answer or returns the ringing.

4 What kind of conductor this can be

The construction does not have a free choice of target, and identifying it correctly settles most of what follows. Non-overlap fixes two things at once, and they happen to be the two parameters of the Hubbard Hamiltonian. The on-site repulsion U — what it costs to put two electrons on one site — is infinite, since two electrons may not share a domain at all; and the transfer integral t — the amplitude to hop to a neighbour — is exactly zero, since it *is* the overlap of neighbouring orbitals and there is none. In that language the construction is the **atomic limit** and not the Mott limit — the reading it is usually given — and the difference is not a refinement:

	$U \to \infty$, $t \neq 0$	$U \to \infty$, $t = 0$
half filling	Mott insulator	isolated atoms
superexchange	$J = 4t^2/U$	none
doped	**conducts**	still nothing
spin dynamics	yes	none

A Mott insulator is insulating but alive: it carries magnetic exchange, and doping it produces a conductor, which is what the cuprates are. The atomic limit is a collection of atoms with electrostatics between them. This accounts for every negative result below, and for one in particular: doping a Mott insulator works *through* t, so at $t = 0$ there is nothing for doping to activate — which is exactly what the measurement shows, the barrier rising with carrier concentration rather than falling. Metals are therefore out of scope in kind and not by degree, their carriers being extended states and their transport band motion, neither of which an arrangement of densities on domains appears able to supply. Two results usually read as failures follow from that scope rather than against it: $d\sigma/dT > 0$ is *correct* for an activated conductor and fatal only for a metal, and the absence of a Fermi surface does not bear on a dilute carrier gas, which is non-degenerate and has none to reproduce.

What remains within reach is matter whose carrier is a localised charge on a site, whose gap is on-site repulsion rather than band structure, and whose transport is activated hopping. That is a real class — doped Mott insulators, organic and molecular semiconductors, polaronic oxides, amorphous semiconductors — and this article set out to claim it. §6 measures the activation energy and finds 613–1723 meV, against the 10–500 meV those materials show — so on that geometry the claim does not survive its own measurement. It survives on another. §9 repeats the measurement on a cable, where the carriers stand off ion cores rather than occupying them, and returns ≈240 meV: inside the class. The scope claim therefore holds for the arrangement conducting matter actually has, and fails for the one in which the carrier sits on its own kernel — which is a statement about which model to use, not about the framework. Silicon and the III–Vs were never in scope, their gaps being band gaps.

5 The interface, and what it does not resist

One property of the free interface conditions everything the partition does, and it is internal to the construction — no comparison with another theory is needed to state it. *A flat density is* *admissible on a free boundary at no cost.* A constant ψ_i satisfies the normalisation, has $\nabla\psi_i = 0$, and minimises the electrostatic energy against a neutralising background, so redrawing an interface between domains of equal density costs nothing at all. That was already used, in §6, as the reason the boundary is free to move; here it is the same fact seen from the other side.

What follows is that the partition offers no resistance to being compressed or rearranged uniformly. A gas of these domains has no pressure of its own: squeeze it and, so long as the density stays uniform, nothing in the energy objects. It also has no resistance worth the name to being *modulated*, since the only penalty on a non-uniform density is the gradient term, and that term carries the coefficient an atom's spectrum sets rather than one an extended arrangement chose.

This is not a comparison with the electron gas of textbook theory, and it is deliberately not offered as one. That model presumes a smeared positive background and delocalised electrons, which is the ontology this framework denies; asking these domains to reproduce its coefficients would be asking them to be something else. The statement here is internal, and it is what §9 turns on: a partition with nothing resisting modulation is a partition whose carriers will bunch into whatever wells a lattice provides.

The interface condition is not fixed by the framework, though. It is a choice — free, Dirichlet, or a Robin condition in between — and a different choice makes a flat density cost something. Which condition matter actually wants is an open question pursued elsewhere; what matters here is that the one used throughout this article supplies no stiffness, and that this is a property of the boundary rather than of the functional.

A caution about specified partitions, learned by getting it wrong. The method of §6 evaluates the energy of a *stated* partition, and it is sound only when the stated partition is one the physics admits. The interface is not a surface to be shaped at will: the wavefunction is a single global field partitioned by labels, continuous across every interface, each domain carrying one unit of charge, and the boundary sits where the densities balance. Freezing a partition removes that last condition. An attempt to measure a bending rigidity by imposing a curvature, freezing it and relaxing the fields gave an energy falling symmetrically and quadratically in the imposed curvature, which reads convincingly as a negative rigidity; but released, a boundary started from the curved configuration returns to the flat one. The flat interface is the balance surface and is stable, and what the frozen scan measured was the energy gained by *dropping the balance condition.* A specified partition gives a legitimate energy difference between configurations the physics admits — as symmetry guarantees for the two in §6 — and a meaningless one along a direction it forbids. Nothing in the numbers distinguishes the two cases; the check is to release the boundary and see whether it stays.

6 Computing with it anyway

Electric conduction is the interaction of the electrons with the ion lattice. It is the elementary case — ordinary matter at ordinary temperature — and it is for that reason the *demanding* one here, since ordinary-temperature conduction is degenerate matter, and degeneracy is what this framework has least of.

One property of the construction is needed before anything else, and it is the property that makes conduction a live question rather than a hopeless one. With the free interface used throughout, a uniform density is admissible on any partition: $\psi_i \equiv |\Omega_i|^{-1/2}$ satisfies the normalisation, minimises the electrostatic energy against a neutralising background, and has $\nabla\psi_i = 0$. Redrawing the boundary therefore costs nothing,

$$\text{a free interface between domains of equal density carries no confinement energy,} \qquad (2)$$

and the boundary is the only object this framework moves — its solver evolves each interface by the local density balance $(\rho_i - \rho_j)/(\rho_i + \rho_j)$, transferring ownership of a cell when the neighbour's density exceeds its own. Domains do not translate; they are redrawn. *A vanishing interface cost is a vanishing cost of transport.* What that same fact does to the equation of state is a separate matter and is taken up in §5, where it is much less welcome.

Metallic conduction is the paradigm of *delocalisation*: the carriers are extended states spanning the crystal, whereas RealQM's electrons are localised by construction, each in its own domain, with non-overlap requiring N carriers to occupy N disjoint regions. The natural minimiser of (1) is a set of compact cells, which describes a Mott insulator more naturally than a conductor. Nor is there an evident mechanism for the metal–insulator distinction, which in standard theory is band filling — two electrons per band per cell — an exclusion-principle count the framework does not have; and with no Fermi surface, the carrier density in $\sigma = ne^2\tau/m$ has no referent.

What does fit naturally is the opposite regime: localised charges hopping between domains by thermally activated jumps, as in semiconductors, disordered systems and small-polaron transport. That needs no Fermi surface and follows from the same thermal motion that supplies the pressure.

This yields a prediction crude enough to be decisive, and it concerns only a sign. Metallic conductivity *falls* with temperature; hopping conductivity *rises*. If RealQM transports charge by hopping rather than band motion it should give $d\sigma/dT > 0$. Within the scope set in §4 that is the *right* answer, activated transport being what a localised-carrier semiconductor does; it would be fatal only for a metal, which is why the target matters. Whether it does has not been computed for electrons, and the pessimistic reading should be held loosely: the same lattice that reproduces alkali lattice constants is present here too.

For *protons* the mechanism has already been demonstrated in this framework. Grotthuss conduction — an excess proton migrating along a hydrogen-bonded chain of water molecules by successive transfers, rather than by bodily motion of any one particle — is hopping transport in its purest form, and it is a standard RealQM calculation: a chain of waters with O–O spacing near 2.65 Å, an excess proton, and the question whether it hops from one molecule to the next. Related transfers, $\mathrm{HF} + \mathrm{H_2O} \to \mathrm{F^-} + \mathrm{H_3O^+}$ and $\mathrm{HCl} + \mathrm{NH_3} \to \mathrm{Cl^-} + \mathrm{NH_4^+}$, are computed the same way.

This matters for the argument rather than merely illustrating it. Charge transport by transfer between localised domains is not a regime the framework must be stretched to reach — it is one it already handles, in the case where the carrier is a proton.

The electron case cannot presently be tested in the same way, and the obstruction is instructive enough to be worth setting out. Two chain calculations were attempted here — five hydrogens with an excess electron, and five hydrogen kernels with one electron missing, the electronic counterparts of the proton wire — and both returned nothing. Neither result means what it appears to.

The cause is not what it first appeared. Kernel proximity assigns the labels only *initially*; thereafter a free-boundary rule evolves each interface from the local density balance and transfers ownership of a cell when the neighbouring domain's density exceeds its own. Partitions therefore do rearrange, and electron transfer is representable in principle. What defeated the attempts here was specific to each: a carrier given no kernel feels no force from the lattice; a bare kernel is given no domain, so a vacancy has nothing to migrate into; and a carrier placed on an occupied site can seize the nuclear region outright, since with free interfaces no confinement cost opposes it. Each is a property of the construction, not of (1).

The deeper cause survives that repair. Conduction is charge in motion, and (1) is an energy minimisation: it returns the equilibrium partition, not a current. Given complete freedom in the labels one would still obtain the ground state, and for a symmetric chain that state places one domain per site by symmetry, with nothing flowing. A minimisation has no time in it.

Which filling the measurements belong to, and which needs the repair. It is worth being exact about this, because it determines how much of the article depends on β. At *half filling* — one electron per site, every domain occupied — the $U \to \infty$ construction forbids conduction outright: there is no vacancy for a carrier to enter, and the insulator is structural rather than dynamical. That is the case the double-occupancy repair is for. But it is not the case measured here. The chains reported above are doped by ionising every second atom, leaving half the sites empty, and every barrier in this article is the cost of a carrier moving into a *vacancy.* Single occupancy handles that without amendment.

So the transport results stand on their own: they require no finite U, no double occupancy and no β. What requires the repair is the undoped, half-filled chain, which is not run here. This is the stronger position of the two — the measurements are independent of the proposal made to extend them — and it locates precisely what the proposal would buy: not better hopping, but conduction at a filling where the construction currently permits none.

What the framework does possess is an asymmetry of dynamics. Nuclei move, under classical equations of motion; electrons relax, to a static minimum. Charge carried by *nuclei* is therefore representable — which is precisely why the Grotthuss calculation works, the carrier there being a proton — and charge carried by electrons is not, because in this scheme electrons have no equation of motion at all. The correct machinery exists elsewhere in the programme, in the time-dependent form where a real density carries a complex current, and a conduction calculation belongs there rather than here.

7 The threshold field, measured

The question this article asks has a form that avoids barriers altogether, and it is the form in which the answer can actually be obtained.

Put the chain in a field. Every electron shifts to the right. If the shift reaches one full lattice spacing, each electron has taken over its neighbour's kernel — and since the electrons are identical and the energy depends on the partition and not on labels, *the configuration is restored.* That restoration is exactly what separates a current from a polarisation:

tilt below the steepest slope of $E(u)$	u saturates: bounded polarisation, no DC current
tilt above it	u runs: the state cycles, charge flows steadily

This *is* the applied-field, free-boundary case, and it is worth saying so plainly: u is the collective coordinate of the free boundaries themselves, the displacement a field would drive, and a uniform field adds nothing to the energy but a term linear in it, $E(u) - NeFu$. The field therefore tilts the landscape and changes nothing else in it, so the landscape can be computed once, without a field, and tilted afterwards. What is given up by doing it this way is the motion *after* the last stationary point is lost — the rate, which the paragraph below shows would be an artifact in any case — and not the threshold, which is exactly where that point is lost.

So the threshold is not an activation energy and not a saddle. It is the **maximum gradient** of the periodic energy $E(u)$, and §10 has measured that curve on the wrapped chain, where $E(u{+}a) = E(u)$ holds to every digit and the three reflection pairs agree to 1–2 $\times$ 10^{-5} Ha. Nothing here differences distant configurations.

From the measured curve at $r_c = 1.93$, $a = 5.5$ a$_0$ — both parameters determined rather than chosen — the steepest segment gives $|dE/du| = 0.0982$ Ha/a$_0$ for seven electrons, and a sinusoidal fit, which recovers the peak a chord underestimates, gives 0.1194. Per electron:

$$F_{\text{dep}} = 0.014\text{–}0.017 \text{ au} = 7\text{–}9 \times 10^{9} \text{ V/m} = 2.1\text{–}2.6 \text{ V per lattice site.} \tag{3}$$

Copper carrying an ampere runs at 4×10^{-12} V per site. The separation is **twelve orders of magnitude**, and it is not a number any refinement moves. Nor is such a field available: dielectric breakdown occurs two to three orders of magnitude below 10^{10} V/m, so the lattice is destroyed long before the electron distribution depins. *The construction does not conduct under any field a material survives.*

Above the threshold, and why the rate there is not a result. Drive harder than (3) and the partition does run: the tilt exceeds every slope of $E(u)$, no configuration is stationary, and the solver will advance the interfaces without stopping. It will also report a speed, and that speed is worth naming as an artifact before anyone quotes it. What sets it is the descent parameter of the relaxation scheme, not the physics — as §3 sets out, the pseudo-time carries no calibration — so the same chain under the same field yields whatever current the numerical mobility is set to. The threshold is a property of the measured curve $E(u)$ and survives refinement; the current above it is a property of the scheme and does not — and the regime is in any case the one just excluded.

A defect in the model behind these numbers, and what it leaves standing. The pseudopotential exclusion used throughout this section keeps valence density out of a domain's *own* core only. A cell belonging to domain A but lying inside neighbouring core B is not excluded, so A's density enters B's core and collects its attraction. That is invisible wherever each domain contains its own core — which is why $d = 0$ is unaffected — and decisive at $d = \frac{1}{2}$, where every core is split between two domains and the exclusion never fires on either half. Excluding from *every* core instead inverts the registry: the minimum moves to $d = 0$, atom-centred, and the bond-centred preference reported here disappears.

Neither treatment is usable as it stands. Own-core exclusion under-excludes and gives the wrong registry; full hard exclusion over-confines, since at $r_c = 1.93$ and $a = 5.5$ the excluded spheres leave only $1.64\, a_0$ between neighbours and the electron at $d = \frac{1}{2}$ becomes a particle in a box rather than a charge in a lattice. The two therefore *bracket* the answer rather than settling it, and what is needed is a soft repulsive core, or r_c recalibrated with full exclusion active. $V_0 = 816 \pm 4$ **meV and the threshold field** (3) **are the own-core values**, verified as such but resting on an exclusion rule that is not the model. Refinement cannot repair this: the mesh convergence below is convergence of the wrong quantity.

The scaling route of §8 does not inherit the defect. Its measurements are made at an undisplaced partition, where each domain contains its own core and the two rules agree to five decimals, so $\alpha = 27.4N$ and the extensivity verdict stand independently of how the exclusion is settled. That the article reaches the same conclusion by two routes is what makes the loss of one of them survivable.

The corrugation is bounded below, at every density. Repeating the scan across lattice spacings from 3.5 to $8.0\, a_0$, with the translation control exact at every one, gives V_0 per electron of 1513, 972, 812, 783 and 816 meV. It has a minimum near the equilibrium spacing and rises on both sides — compressed, the cores crowd the electron; expanded, sharing a kernel costs more than owning one — so across a factor of 2.3 in lattice constant the corrugation never approaches zero and the registry preference never flips. There is no free-sliding point at any density. These are own-core values and inherit the bracket above, but the qualitative statement does not turn on the exclusion rule: a vanishing V_0 at some spacing is the one escape the extensivity argument of §8 leaves open, and no spacing takes it.

What is bounded, and what is not. Everything measured here is the *rigid* slide: all N electrons displaced together, the spacing held fixed, the pattern translating without deforming. That is the expensive path, and its cost grows with chain length. The cheap path is a kink — a single carrier advancing while the rest hold still, compressing the domains ahead and expanding those behind — whose cost is length-independent, and that is the configuration real conductors use. *The kink barrier is not measured in this article.* On the cyclic geometry it requires the slip to accumulate to exactly one unit around the loop, $f(j + N) = f(j) + 1$, which the profile used here does not satisfy; the attempt misses its own control by 997 meV and is not reported. Note also that the maximally delocalised limit of such a profile gives evenly spaced walls and a barrier of exactly zero, so the kink barrier is a function of its localisation and the physical value is the minimum over that — a two-parameter determination, not a single number. A second qualification points the same way: the chain is unstable to non-uniform distortion, a sinusoidal displacement *lowering* the energy, because a domain here cannot change position without changing width and a wider domain lets its electron spread. So (3) is an upper bound on two counts, and neither is remotely capable of closing twelve orders of magnitude.

And the largest qualification of all. Every energy in this article is computed with the **nuclei clamped** at their lattice sites. So $V_0 = 812$ meV is the rigid-lattice corrugation, and (3) the rigid-lattice threshold — both upper bounds, because in real matter the lattice relaxes around the carrier and that relaxation is the dominant term. It is what makes small polarons work at all: the Marcus–Holstein barrier is $E_a = \lambda/4$ with λ the reorganisation energy, and $\lambda \approx 0.1$–0.3 eV in organic semiconductors gives 25–75 meV, far below the bare electronic corrugation of those same materials. Nothing here tests it, and the framework is not prevented from doing so — nuclei move in this scheme under classical equations of motion, which is the half of the dynamics it possesses and the reason the Grotthuss calculation works. If lattice relaxation reduced the corrugation by the factor it does in polaronic materials, 812 meV would fall toward 100 meV and the threshold with it, from $\sim$8 to $\sim$1 GV/m: still above dielectric breakdown, but no longer by three orders. This is the one accessible variation that could move the verdict rather than confirm it, and it is untested.

The two bounds are independent and they multiply, which is worth stating in one place rather than leaving to be assembled: a clamped lattice and a rigid slide both raise the number reported, the first by an estimated order of magnitude and the second by a factor nobody here can quote, since the kink barrier is the quantity §10 shows cannot be extracted. Twelve orders is therefore the separation for the mode measured, not a floor under every mode. What survives either way is the sign of the conclusion — the field required is far above the one at which a wire carries current, and the reduction would have to reach nine orders to change it.

8 The same verdict from scaling

The measurements in this section are on a *different chain* from §7, and that should be said before any of its numbers appear. It is a model atom of nuclear charge +1 with core radius $r_c = 1.0\,a_0$ and one valence electron, seven of them in a row at spacings from 3.5 to $8.0\,a_0$ — parameters chosen to define a system rather than to represent an element. The row is an arrangement of kernels, not a property of the construction: the domains are regions of ordinary three-dimensional space, and the same computation runs unchanged on a square or cubic lattice of the same atoms. Its corrugation runs from 182 to 461 meV per electron across that range, and is 379 meV at $a = 4.5\,a_0$, the spacing at which the polarisabilities below are computed. Nothing here is a second determination of the quantities of §7; what is claimed is a statement about *scaling*, which is independent of which chain it is made on.

That independence is worth making concrete, since the natural objection is that the verdict rests on two chains rather than one. Extensivity does not involve r_c at all: (5) is proportional to N for *any* positive stiffness, so r_c can move α_{intra} and K — both per-atom constants — without touching the power of N. What r_c could do is take the one escape, $K \to 0$, and the chain of §7 is measured to be further from it than this one: $V_0 = 812$ against 379 meV per electron, hence

$K \simeq 530$ against 369 meV/a$_0^2$ and a slide term $e^2/K \simeq 51$ against $73.6\,a_0^3$. The stiffer chain is the safer one. Recomputing α at $r_c = 1.93$ would therefore move the coefficient 27.4, which is reported as a measurement of this chain and not as a property of the framework, and would leave the classification where it stands.

The threshold field settles the question by driving the pattern. It can also be settled without driving anything, by asking how the response grows with the size of the sample — an independent route, resting on different assumptions, and worth having because the two can be checked against each other.

One point about what is being asked deserves stating before any of it, because it decides which part of the measurement carries the weight. An isolated finite system cannot conduct, whatever it is made of. Place a piece of copper in a box and apply a static field: the charge redistributes to the surfaces, the interior field cancels, and everything stops. Copper polarises and stops. So the fact that a response *saturates* is never evidence of insulation — it is what any closed system does, and no calculation on a bounded sample can conclude otherwise.

What distinguishes the two cases is not whether the response saturates but how the saturated value *scales*. A metal's polarisability grows faster than the number of atoms; an insulator's is extensive. That is what Kohn's argument supplies and what makes a closed, static calculation capable of answering the question at all. The measurement that carries the classification here is therefore $\alpha = 27.4N$ — the extensivity, not the boundedness — and the corrugation enters only because a vanishing K is the one way extensivity could fail.

8.1 Why the scaling is fixed before it is measured

Nor is the outcome fixed in advance by the postulate. At full filling the domains tile space, but transport does not require a vacancy: every boundary may shift at once, translating the whole partition. That collective mode has a polarisability. Writing K for the per-electron curvature of the corrugation the partition slides through,

$$E(u) = \tfrac{1}{2} K_{\text{tot}} u^2 - NeFu \quad \Longrightarrow \quad \alpha_{\text{slide}} = \frac{N^2 e^2}{K_{\text{tot}}} = \frac{Ne^2}{K}, \tag{4}$$

since the stiffness is extensive, $K_{\text{tot}} = NK$.

Each domain carries exactly one electron. The constraint $\int_{\Omega_i} \psi_i^2 = 1$ is imposed, not obtained, so the monopole of every domain is pinned and **no charge passes between sites**. The sites, however, are not the fixed points of the problem — the kernels are. A domain may arrive at the kernel that was its neighbour's while carrying the same unit of charge it began with, so transport here is *the sites moving*, not charge moving between them, which is the dislocation relation again and the reason nothing needs to hop. That removes the mechanism which gives extended systems their large polarisability, and leaves exactly two responses to a field: the density deforming within each fixed domain, and the domain boundaries displacing. Write α_{intra} for the first — the polarisability an atom would have with its domain held rigid, the electron leaning toward the field while its walls stay put — and use (4) for the second:

$$\alpha = N\Big(\alpha_{\text{intra}} + \frac{e^2}{K}\Big), \tag{5}$$

which is proportional to N exactly. Allowing the walls to move differentially does not change this: with elastic coupling κ between walls and substrate stiffness K per site, the displacement profile satisfies $\kappa u'' - Ku = -f$ and returns $u \simeq f/K$ through the bulk with corrections over a length $\sqrt{\kappa/K}$ at the ends, so the total dipole remains proportional to N.

The one escape is $K \to 0$. Only when the corrugation vanishes is the displacement limited by the chain length rather than by a restoring force, and only then does α grow faster than N.

8.2 Polarisability against chain length

Computed for a model chain — one-valence-electron atoms of core radius $r_c = 1.0\,a_0$ at spacing $a = 4.5\,a_0$, frozen partition, from the induced dipole at two field strengths well below the depinning threshold:

N	α	per atom	linearity
3	81.57	27.19	4.3%
5	137.26	27.45	5.5%
7	192.50	27.50	4.8%

Table 1: Polarisability against chain length. The last column is the spread between the two field strengths. It is the precision of the measurement rather than a departure from linearity: the fields are already at a tenth of the depinning threshold, and lowering them further puts the induced energy below the solver's own convergence noise, so 5% is the floor available here.

The polarisability per atom is constant to 1% across the three lengths, so

$$\alpha = 27.4\,N, \tag{6}$$

reproducing all three entries to better than 1% with no intercept required. The response is extensive, with $\alpha_{\text{bulk}} = 27.4\,a_0^3$ per atom.

A surface term is not resolved. At the 5% precision of these values an intercept anywhere between roughly ± 10 is admissible, so the ends of the chain cannot be shown to differ from its interior — which is consistent with the derivation, since a constraint that fixes the charge on every domain makes no distinction between an end site and a middle one, but it should be read as an absence of evidence rather than evidence of absence.

This is (5) with its coefficient supplied, and it confirms rather than tests the derivation: extensivity follows from the postulate, and what the measurement adds is the size of the coefficient and the fact that nothing anomalous happens as the chain lengthens.

One caution belongs with that coefficient, and it is not small. The partition is frozen, so e^2/K drops out of (5) entirely and what is measured is α_{intra} alone. Taking K from the corrugation at the same spacing gives $e^2/K = 73.6\,a_0^3$, nearly three times α_{bulk}. Releasing the partition would therefore roughly triple the response: the wall displacement is the *larger* part of it, and the numbers above are a lower bound on the polarisability rather than an estimate of it.

It was frozen for a reason of resolution and not of convenience. The solver moves an interface by transferring ownership of whole cells, so a wall sits at a cell boundary, while a sub-threshold field asks it to move by a fraction of one: it stays put, or jumps the full cell. A released measurement returns a staircase rather than a linear response. That is also why §7 imposes u rigidly instead of letting a field find it — an imposed displacement can be set to any value and checked exactly, since translation by one spacing must return the same energy. Releasing the partition and re-measuring α directly, which would test the factor of three above, needs a sub-cell representation of the interface and has not been done.

8.3 What the boundary treatment fixes, and what it does not

V_0 is a difference between two *wall placements*, so of every quantity reported here it is the one most exposed to how boundaries enter the kinetic energy. That exposure was measured rather than assumed. Reassembling the same converged states with the surface condition changed — the default Neumann at walls and cores, then Dirichlet at the walls, then Dirichlet at the cores — gives +379, +721 and −949 meV per electron at $a = 4.5\,a_0$.

Two readings follow, and they point opposite ways. The corrugation is *surface-dominated*: the bulk kinetic energy is nearly unchanged while V_0 moves by 1300 meV and reverses, which is direct support for the steric character argued below, since a quantity fixed at the exclusion surfaces is by definition not an electronic one. But it also means neither the magnitude nor the sign of V_0 can be read as a property of the model alone. The Dirichlet variants are a stress test rather than a rival physics — forcing the density to vanish across one cell adds a large and mesh-dependent surface term — yet the sensitivity they expose is real.

What survives is what the classification needs. $|V_0|$ is 379, 721 and 949 meV in the three treatments; it approaches zero in none of them, so K stays finite and the polarisation bounded however the surfaces are handled. The verdict is robust where the numbers are not.

8.4 Two routes, one verdict

These are different chains, as stated at the head of the section, so the agreement claimed is qualitative — that both give a bounded response and a positive curvature — and not a numerical coincidence between two determinations of one quantity.

The threshold field of §7 and the extensivity measured here are independent. One drives the partition and reports the field at which it would run; the other never applies a field large enough to move anything and reports how the bounded response scales. They rest on different assumptions — the first on a periodic energy curve and its steepest gradient, the second on the Kohn–Resta criterion and the growth of α with N — and they agree: the polarisation is bounded, the curvature is positive, and there is no DC current. That agreement is the strongest statement this article makes about the classification, because neither measurement could have produced it alone.

9 Two models: the atomic chain and the ion–electron cable

The measurements above are all on one arrangement: a chain of atoms with the valence electron occupying the same region as its kernel. That is a real system — a hydrogen chain is exactly this — but it is not how conducting matter is built, and the verdict turns out to depend on the difference.

What changes, and why it should. In an ion lattice the carriers do not sit on the nuclei; the core electrons do, and the conduction charge stands outside them. Representing that needs two populations: cores held inside a radius, carriers outside it, with only the carriers free to slide. It is the one arrangement that gives a carrier a standoff *without* violating the tiling, because the inner region is genuinely owned — by the core electrons — rather than left empty. In that geometry the ion presents its net charge to the carrier from a distance, and the lattice harmonic the carrier feels is suppressed by $e^{-2\pi r/a}$.

The measurement. A finite chain has ends, and on this construction they dominate: the terminal domains bind an electron-volt harder than the interior, and a corrugation read from total energies is reading them. Displacing only the middle M carriers repairs this. The energy difference is then M times the bulk cost plus two junction costs, so differencing two window sizes cancels the junctions exactly and no terminal domain moves in either configuration:

$$\Delta E(M) = M\,\varepsilon_{\text{bulk}} + 2\varepsilon_{\text{junc}}, \qquad \varepsilon_{\text{bulk}} = \frac{\Delta E(M_2) - \Delta E(M_1)}{M_2 - M_1}. \tag{7}$$

The decomposition asserts linearity in M, which a third window tests rather than fits.

What moves is the same object throughout. Nothing in this section changes the carrier. A domain holds one unit of charge by constraint, so no charge crosses between sites; what advances is the *partition*, and it advances by its boundaries being redrawn. The free boundary is the carrier in all three arrangements below — the dislocation relation of §12, in which each carrier steps forward one spacing as the disturbance passes and charge arrives at the far end while nothing material traverses the sample. The three regimes are not three mechanisms. They are three heights of the barrier that one mechanism has to cross.

The two models, and the range the second covers. With the ion lattice, the standoff and the mesh held fixed:

model	corrugation per carrier	regime
atomic chain: carrier on its own kernel	$\sim$1 eV	insulator; outside the hopping class
cable: carriers standing off ion cores	$\approx$240 meV	semiconductor range: activated hopping
the same cable, carriers stiffened	$\approx$24 meV	metallic onset: unpinned

The first two rows are two *models*: the same framework and the same interaction, asked about two different physical situations — an atom's electron on its own kernel, and a conduction charge standing off an ion core. The third row is not a third model but the same cable with one property varied, and it is the informative one. Doubling the carriers' resistance to being modulated — with the ions, the standoff and the mesh unchanged — drops the corrugation by a factor of eleven. So the pinning is set by whether the carriers bunch into the wells, not by the strength of the lattice, and the question becomes what supplies that resistance.

Where the coefficient comes from, and what it is entitled to be. The $\frac{1}{2}$ in $-\frac{1}{2}\nabla^2$ is not a free choice: it is fixed by calibrating the framework against atomic spectra, hydrogen first among them. It is standardised *there*, on a bound electron in a single potential well. An extended system is a different object — as an effective mass in a solid differs from the free-electron mass without anyone regarding it as tampering — and the coefficient that governs how a carrier lattice resists being modulated by an ion lattice has no obligation to equal the one an atom's spectrum sets. Raising it is therefore not the introduction of a parameter but a question: what stiffness does an extended arrangement actually carry? A degenerate electron gas resists density modulation because compressing it costs Fermi energy. Here a flat density is admissible on a free boundary at no cost, so the coefficient in $t(n) = C\,n^{5/3}$ is $C = 0$ exactly — §5, where it was recorded as a defect of the equation of state. It is the same missing stiffness, and §5 says where it is missing from: the free interface admits a flat density at no cost, so the partition resists neither uniform compression nor modulation, and the only penalty on a modulated density is a gradient term carrying the coefficient an atom's spectrum sets. Supply that resistance and the pinning goes. Nothing here requires comparing the framework with an electron gas; the statement is about its own interface. And §5 also names where the repair would go — the interface condition is a choice, and one that makes a flat density cost something would supply what is missing — so the prediction is sharp: a RealQM whose interface resists modulation should not insulate here.

What this delivers, and what it does not. In the activated regime the corrugation is the activation energy, and the conductivity of a hopping conductor is $\sigma \propto \nu \exp(-E_a/kT)$ — so the table supplies the factor that decides the temperature dependence and the ratio between regimes, 240 against 24 meV being four orders of magnitude at room temperature. What is missing is the prefactor ν, an attempt frequency, which is a *lattice* quantity: the carrier sits in a well and tries once per vibrational period, so $\nu \simeq (1/2\pi)\sqrt{K/M}$ with K the curvature of the same landscape measured here and M an ionic mass. Both are available in this framework — it computes forces on kernels and moves nuclei classically — so an absolute hopping conductivity is a phonon calculation away rather than a missing piece of theory. It is not attempted here.

The metallic case is different and is genuinely outside. There $\sigma = ne^2\tau/m$ and τ is a momentum relaxation time set by electron–phonon scattering, which needs a current as an object, electron dynamics and irreversibility. A minimisation over real densities has none of the three. So the third row says the pinning can be removed; it does not say a metallic conductivity can be computed.

Status. The corrugations above are provisional in a specific way that should be stated rather than buried. They are rigid-slide values, hence upper bounds — a kink depins lower. They come from states whose Euler–Lagrange residual is 2–3 $\times 10^{-2}$, not the 4×10^{-3} the chain reaches. The stiffened runs carry a halved timestep, so they are less relaxed than the others, and the factor of eleven rests on two coefficients rather than a curve. What is not provisional is the instrument: (7) cancels ends and junctions by construction and verifies itself against a third window.

10 The route that did not work, and what it cost to find out

The measurement of §9 was not the first attempt. The first asked for the barrier directly — the energy difference between a carrier here and a carrier one site along — and it cannot be obtained at any resolution this or any comparable calculation will reach. The reason is the scale problem of §1 in its sharpest form: the quantity is about 0.01 Ha, the totals it must be differenced from are 1.3 Ha, and the first-order discretisation error on those totals is 0.27 Ha. Reaching ten per cent of the barrier would need a $42\,000^3$ grid. Brute force does not help, because the error does not cancel between two configurations that differ in more than registry.

The control that showed it. Scanning a full lattice period instead of sampling two configurations supplies the diagnosis for free: translation by one spacing must return the same energy. Six of eight such scans failed that identity, the two that passed survived no change of parameter, and the scan at parameters the framework itself fixes failed by the widest margin. What the scans did contain, reproducibly, was a drift of the carrier toward a chain end and grid-locking of a staircase interface — both of order an electron-volt, which is why they were mistaken for signal for as long as they were.

Earlier figures, not quoted. A compression test on a ring, a doping series, a lattice barrier of order 5 eV between interstitial pockets, and an interface defect at 179 meV independent of chain length. Each specified or evolved a partition that no symmetry fixes, so each is a bound of its stated order rather than a determination. They are superseded by the measurements above and are recorded here only so that the earlier numbers are not read as results.

What this says about scope. Narrower than it first appears. That this barrier is not extractable does not leave the scope claim unmeasured: §9 obtains it by another route, on a geometry with a symmetry that fixes the partition and by a differencing that removes the ends. What this section establishes is that the quantity cannot be reached by comparing configurations far apart in the landscape — and that the structural part of the claim, non-overlap and a vanishing transfer integral, needs no grid at all.

The lesson, which is the transferable part. Ask for a quantity in the form the discretisation can answer. A difference between distant configurations inherits the full discretisation error of both; a difference between neighbouring ones, on a geometry whose symmetry fixes the partition, does not. §7 asks for the slope of a curve sampled a fraction of a spacing apart, and §9 asks for an increment between two window sizes with the ends held still. Both are the same physics asked in a form that survives being computed.

A repair that is not available. An obvious response to a barrier is to let valence electrons share a territory, which removes it by construction. The $n^{5/3}$ scaling rules that out: sharing destroys the extensivity the partition supplies, and a metal would lose its kinetic energy density by fifteen orders of magnitude.

11 The carrier itself, against exact diagonalisation

Everything above is internal to the framework. One external check is worth keeping, and the sharpest form of it does not compare energies at all. Take the same object this article hops — n protons, $n-1$ electrons, nuclei clamped — and diagonalise it exactly in a minimal basis. The Loewdin site populations then say directly whether the carrier is localised:

a (a_0)	E_{corr} (eV)	$4t$ (eV)	max hole/site	
3.0	−6.43	11.11	0.223	delocalised
4.0	−11.97	4.65	0.215	delocalised
4.5	−14.64	2.96	0.226	delocalised
6.0	−19.86	0.71	0.253	delocalised

A hole spread evenly over seven sites reads 0.143; one sitting on a single site reads 1.000. At every spacing the ground state puts it on four or five. *There is no localised carrier, and therefore no barrier for one to cross.* Not because correlation is weak — E_{corr} reaches 19.9 eV — but because strong correlation and localisation are different things: a one-dimensional Mott insulator doped with a single hole conducts, the holon propagating freely.

The disagreement is therefore about the *state* and not about a number, which is the harder kind to explain away. Two qualifications keep it honest: a minimal basis is exact diagonalisation within a deliberately crude span — STO-6G misses the hydrogen atom by six per cent before any chemistry begins — and larger bases did not converge for the stretched chains, so no exact reference exists for these configurations and nothing here declares the pinning wrong for them. What it does establish is the direction of the error. The construction over-localises, by construction, being $U \to \infty$: it fails as §4 said it would rather than arbitrarily, and where the mechanism of this article would actually be tested — a doped Mott insulator, an organic semiconductor, a polaronic oxide, where on-site repulsion and not band structure sets the gap — no comparison is made here.

Two controls added after the fact, and what each is blind to. The first is free and should have been applied from the start: *domains related by a symmetry of the configuration must have equal eigenvalues.* Each domain's stationarity condition carries an ε_i, and on a chain where every site has the same environment they must coincide. A total energy is one number and can look reasonable while the state beneath it is inhomogeneous; the eigenvalues expose that directly. Applied to the coaxial geometry of the follow-on work it showed terminal domains binding an electron-volt harder than interior ones — which identified a corrugation as an end artifact rather than a registry effect — and, on longer chains, a smooth bowl across the whole chain that distinguishes a slowly converging one-dimensional Coulomb sum from a local end effect. It is blind to exactly what the mesh ladder and the translation control are blind to: an error respecting the lattice symmetry passes all three, which is how the exclusion rule above survived them.

The second is a way to measure a bulk quantity on a chain that has ends. Displace only the middle M carriers by half a spacing: the energy difference is M times the bulk cost plus two junction costs, so differencing two window sizes cancels the junctions exactly and no terminal domain moves in either configuration. The decomposition asserts linearity in M, which a third window tests rather than fits. It is the instrument the rigid slide of a finite chain should have used from the beginning.

A note on what was tested numerically

The plasma frequency $\omega_p^2 = 4\pi n$, the screening relation and the vanishing cost of a flat density on a free interface are analytic. The alkali lattice constants are a radial Wigner–Seitz calculation with the stated pseudopotential and correlation functional, run for five alkalis; the standard column reproducing the series is what makes the comparison meaningful, and its own 44% mean error in the bulk modulus sets the accuracy available at this level of model. Three attempts are reported as null rather than omitted. The interface parameter β was tested on helium at three meshes at fixed imaginary time, giving shifts of $+0.151$, -0.018 and -0.184 Ha as the mesh refined — swinging through zero and growing, with absolute energies moving away from the exact -2.9037. The cause is the geometry rather than the arithmetic: in helium the two domains meet at a plane *through the nucleus*, so interface error and Coulomb-cusp error are superimposed and cannot be separated on a uniform grid. The measurement wants H_2, where the domains meet at the midplane *between* the nuclei in a smooth region. An interface-condition test on helium was not converged, giving a sensitivity that reversed sign under mesh refinement, and no conclusion is drawn from it. Two chain calculations — five hydrogens with an excess electron, and five hydrogen kernels with one electron missing — returned no transport, but for a reason established by reading the constructions rather than by the runs: a carrier without a kernel feels no lattice force, and a bare kernel receives no domain for a vacancy to occupy. Both are recorded because the null results located those obstructions. A caution on the numerics belongs with them. Several attempts returned energies below the physical floor for their particle content, but these track the boundary speed and the seeding rather than the configuration: with the interface driven faster than the densities can follow, or with a carrier seeded directly onto an occupied kernel, the relaxation runs away, while the same configurations at a quarter of that speed settled to physical values. A two-domain shell on one nucleus is in fact stable under the density-balance rule — compressing the inner domain raises its density at the interface and lowers the outer's, which pushes the boundary back out — and both domains carry real kinetic energy there, since the nucleus shapes their profiles. The vanishing of C is a statement about a *uniform* density and does not transfer to a shaped one. Scripts and pages accompany the source.

12 Conclusion

The question was whether a construction of non-overlapping unit charge densities, meeting at free boundaries, conducts. It comes in the two tiers of §3, and the formulation answers one of them.

The **classification** it settles, and settles from a static calculation, which is what Kohn's result licenses: the polarisation is bounded, the curvature positive, the threshold field beyond dielectric breakdown, and α extensive in the number of atoms. This is an insulator, and not by a near miss. What cannot be reached is the **rate** — a conductivity, or the activation energy that would set one — and the reasons for that are worth more than the number would have been.

The carrier is a free boundary, and that identification holds. An electron here is a unit of charge density on a domain; what exists is the partition of space. Transport is therefore the partition pattern travelling, in the relation a dislocation bears to slip: charge arrives at the far end while nothing material crosses the sample. That picture accounts for every null result reported above — each had imposed a carrier obliged to translate bodily through occupied space — and it is why the one transport calculation this framework performs successfully is Grotthuss conduction, where the carrier is a proton and the framework does have an equation of motion for it.

The picture has a definite limit, and it should be stated with the claim rather than after it. Defect-mediated transport is real — dislocation glide, charge-density-wave sliding and ionic conduction all proceed that way — but it is not how a metal conducts. Residual resistivity *falls* as defects are removed, so a carrier that must migrate as a defect predicts the opposite of what purity does. The free boundary is therefore the mechanism this construction is restricted to, not the carrier of ordinary conduction, and its prominence here is a fact about the formalism rather than a discovery about charge transport.

Costing that motion defeats the method, and the control is what shows it. A barrier of 0.01 Ha differenced from totals of 1.3 Ha carrying 0.27 Ha of first-order discretisation error requires a cancellation it does not get, and brute force cannot supply it: reaching ten per cent of the barrier needs a 42000^3 grid. Scanning a full lattice period instead of sampling two configurations supplies the diagnosis for free, since translation by one spacing must return the same energy. Six of eight such scans fail that control, the two that pass survive no change of parameter, and the only scan at parameters the framework does not choose fails by the widest margin. What the scans do contain is a drift of the carrier toward a chain end and grid-locking of a staircase interface — both of order an electron-volt, both exactly reproducible, which is why they were mistaken for signal for as long as they were.

But the question does not require that quantity. Asked as a threshold field rather than as a barrier — and on a cyclic chain, where translation by one spacing is an exact relabelling and the control holds by construction — it becomes the maximum gradient of a measured periodic curve, and §7 gives $7\text{–}9\times10^9$ V/m, twelve orders above the field at which copper carries an ampere and two to three orders above dielectric breakdown. The same curve yields a pinning potential of 812 meV per electron and a positive curvature, hence bounded polarisation and no DC current. The verdict the article set out to reach is therefore available after all; what was never available was the number it first tried to reach it with.

Beneath that lies a structural obstacle no refinement touches. Non-overlap makes the transfer integral vanish identically, so band conduction is absent in kind: there are no extended states to carry it. And in this scheme nuclei move under classical equations of motion while electrons relax to a static minimum — so there is no electron dynamics, and hence no object in the theory corresponding to a current. *A minimisation has no time in it.* Every attempt above infers transport from an energy landscape, which is a proxy, and it is the proxy that breaks. The missing ingredient is not a better barrier calculation but an equation of motion, and it exists elsewhere in the programme: the time-dependent form in which a real density carries a complex current is where a conduction calculation belongs — and it supplies the current, not its rate, which needs the dissipation no framework derives unaided.

What the framework is, and is not. Single occupancy with long-range Coulomb is *not* the Hubbard atomic limit, whose degeneracy over carrier position would give a barrier of exactly zero; the Coulomb interaction between extended densities breaks that degeneracy, placing the construction closer to an extended Hubbard model where charge ordering is genuine physics. Within bound matter — structure, binding, geometry — it works, and the transport regimes it reaches are those with nuclear carriers. An electronic transport *rate* is outside, and against exact diagonalisation on the same chains the reason is visible directly: the exact carrier occupies four or five sites where this construction admits only one.

On the system tested. Every measurement reported here is on a regular chain of atoms carrying one valence electron each. The extension to a regular two- or three-dimensional lattice needs nothing new — the machinery is three-dimensional already and only the kernel positions change — but it has not been made, and a partition with more neighbours to slide against is where the corrugation would next be tested.

What the time-dependent form would have to supply. That the framework is static here is a fact about this article, not about RealQM, and the route out is specific enough to state. What has to be supplied is a velocity, and it need not arrive as a complex phase: a real density carrying a real velocity field is Madelung's form of the same equation [4], and in this framework the natural variables are the densities together with the velocities of the boundaries that bound them, with $J = \rho v$ and continuity. Either way a current exists as an object, and the first obstacle goes. It does not remove the second, since a Hamiltonian dynamics under a constant field rings rather than settles, exactly as the $3N$ equation does, and a steady current needs dissipation. But the dissipation is already in the scheme and need not be imported: the nuclei move under classical equations of motion, so energy can leave the electrons into lattice motion, and the electron–lattice channel that §3 found every treatment inserting by hand — a collision term, a partial trace, thermalising reservoirs — would here be present in the equations being solved. A resistivity obtained that way would have τ as an output. That is the strongest form the claim of this programme could take on transport, and nothing in this article establishes it. What time-dependence would *not* change is the exclusion: a phase does not make domains overlap, so there is still no transfer integral and no extended states, and the target remains activated transport with localised carriers.

The verdict depends on the arrangement, which is the article's main change of view. Measured on a chain where the carrier occupies its own kernel's region, this framework insulates and falls outside even the activated-hopping class it claims. Measured on a wire — cores inside, carriers standing off them, which is how conducting matter is built — the corrugation drops into that class. And given the stiffness against density modulation that a degenerate gas has, it drops to thermal energy and the pinning goes. So the insulating verdict of §7 is a verdict about a geometry, not about the framework, and the framework's real obstruction is the one §5 identifies: the free interface leaves the partition with nothing resisting modulation. That the same missing term should produce both a wrong compressibility and an insulator is the most useful thing this article has to report, and it comes with its own repair — the Robin interface at $\beta a = 1.146$ — which makes it a prediction rather than a diagnosis.

The result is therefore a boundary, not a barrier. A method with a mapped edge is more usable than one claimed to be universal, and the edge is worth stating exactly rather than sweepingly, since its four parts are not of the same kind. *Metallic conduction* is excluded structurally: non-overlap leaves no extended states and no transfer integral, and no refinement produces them. A *conductivity* is unavailable in the static form, which has no phase, hence no current as an object and no clock to give one a rate. *Nuclear* transport is inside, protonic conduction being the one transport calculation that works. And electronic transport by boundary motion is *described rather than forbidden*: the partition slides, the free boundary carries the charge, and what this article reports is not the absence of that mechanism but its pinning — on one chain, in the rigid mode, on a clamped lattice, with two upper bounds that both push the number down. A pinned mechanism is a described one. Everything this article originally set out to measure — a carrier barrier, a hopping rate — lay on the far side of the first two.

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